

VALIDATION REPORT
OPERATIONAL QUALIFICATION
CLINICAL MODULE

JR Validated Environment — Clinical Module v5.0
Module 1: Study Design | Module 2: Diagnostic Accuracy | Module 3: Survival
Analysis

Field	Value
Document Number	JR-VR-CLIN-001
Title	Validation Report — Operational Qualification, Clinical Module
Validation Plan	JR-VP-CLIN-001 v5.0 (Rev 5)
System	JR Validated Environment — Clinical Module
Module Version	5.0
Document Version	5.0 (Rev 5)
Execution Date	2026-07-20
OQ Result	PASS — 196 / 196 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-07-15	Joep Rous	Initial OQ execution report — Module 1 (Clinical Study Design). 36 / 36 test cases passed.
2.0	2026-07-16	Joep Rous	Scope extended to Module 2 — Diagnostic Accuracy (jrc_clinical_dx_accuracy, jrc_clinical_dx_ss, jrc_clinical_dx_roc). Re-execution of the full clinical suite: 87 / 87 test cases passed.
3.0	2026-07-17	Joep Rous	Scope extended to Module 3 — Survival Analysis (jrc_clinical_km, jrc_clinical_coxph). Re-execution of the full clinical suite: 123 / 123 test cases passed.
4.0	2026-07-19	Joep Rous	Module 2 (Diagnostic Accuracy) extended with jrc_clinical_dx_compare (paired AUC comparison, DeLong 1988). Re-execution of the full clinical suite: 141 / 141 test cases passed.
5.0	2026-07-20	Joep Rous	Scope extended to Module 4 — Comparative Analysis (jrc_clinical_compare_means, jrc_clinical_compare_props, jrc_clinical_ancova, jrc_clinical_logistic). Re-execution of the full clinical suite: 196 / 196 test cases passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-07-20
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Clinical Module, version 1.0. It provides objective evidence that all clinical study design scripts perform as specified in the OQ Validation Plan JR-VP-CLIN-001 v5.0 under the defined test conditions.

This report records the test environment, execution results for all 196 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by `admin_clinical_oq`.

2. Scope

2.1 In Scope

- All 3 R scripts in `repos/clinical/R/` (`jrc_clinical_ss_means`, `jrc_clinical_ss_props`, `jrc_clinical_ss_survival`)
- Two-arm parallel-group sample sizes for superiority, non-inferiority, and equivalence (TOST) frameworks (survival: superiority and non-inferiority only)
- Numeric correctness against independently derived references (Chow, Shao & Wang 2008 ch. 3/4 normal approximations; Schoenfeld 1983 events formula)
- Framework-specific alpha and sidedness reported with the implied z-value
- Allocation ratio, dropout inflation, and sensitivity scenario tables computed in the validated R layer
- Correct rejection of invalid or incomplete designs with informative error messages
- Bypass-protection verification (`RENV_PATHS_ROOT` check)

2.2 Out of Scope

- Infrastructure scripts in `bin/` and `admin/` (covered by JR-IQ-001)
- The Streamlit GUI wizard (`app/clinical_design.py`) — pure orchestration; all figures come from the scripts validated here
- Performance qualification (PQ) with real study protocols
- Validation of the R interpreter
- Statistical validation of the published methods themselves (Chow, Shao & Wang 2008; Schoenfeld 1983 — referenced in JR-VP-CLIN-001)

3. Reference Documents

Document ID	Title	Location
JR-VP-CLIN-001 v5.0	Validation Plan — OQ, Clinical Module	<code>repos/clinical/docs/clinical_validation_plan.pdf</code>
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite	<code>docs/oq_validation_plan.docx</code>
JR-VP-001 v1.0	Validation Plan — Installation Qualification	<code>docs/</code>
JR-IQ-001	IQ Execution Evidence	<code>docs/iq_validation_20260311_205146.txt</code>
OQ Evidence	Clinical OQ Execution Evidence (pytest output + header)	<code>clinical_oq_execution_20260720T104013.txt</code>

4. Test Environment

Component	Details
Execution date / time	2026-07-20T10:40:14
Host	Joeps-MacBook-Air.local
Operating System	[not found]
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/clinical/admin_clinical_oq
Test directory	repos/clinical/oq/
Test data	(none — the scripts are parameter-only)
Execution command	bash repos/clinical/admin_clinical_oq
OQ evidence file	clinical_oq_execution_20260720T104013.txt

5. Test Data

The clinical study design scripts are parameter-only: a sample size is computed before any study data exist, so every test case drives a script through jrrun with command-line flags alone. No CSV fixtures are used. The numeric reference values asserted by the test cases are derived independently of the scripts (documented per test case in the OQ test docstrings and in JR-VP-CLIN-001).

6. Test Results

All 196 test cases were executed on 2026-07-20. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

6.1 Summary

Script	Tests Executed	Result
jrc_clinical_ss_means	14	14/14 PASS
jrc_clinical_ss_props	11	11/11 PASS
jrc_clinical_ss_survival	11	11/11 PASS
jrc_clinical_dx_accuracy	20	20/20 PASS
jrc_clinical_dx_ss	14	14/14 PASS
jrc_clinical_dx_roc	17	17/17 PASS
TOTAL	196	196/196 PASS

6.2 Results by Test File

Test File	TCs	Result
test_clinical_ss_means.py	TC-CLIN-MEANS-001..014	14/14 PASS
test_clinical_ss_props.py	TC-CLIN-PROPS-001..011	11/11 PASS
test_clinical_ss_survival.py	TC-CLIN-SURV-001..011	11/11 PASS
test_clinical_dx_accuracy.py	TC-CLIN-DXACC-001..020	20/20 PASS
test_clinical_dx_ss.py	TC-CLIN-DXSS-001..014	14/14 PASS
test_clinical_dx_roc.py	TC-CLIN-DXROC-001..017	17/17 PASS
test_clinical_km.py	TC-CLIN-KM-001..019	19/19 PASS
test_clinical_coxph.py	TC-CLIN-COX-001..017	17/17 PASS
test_clinical_dx_compare.py	TC-CLIN-DXCMP-001..018	18/18 PASS
test_clinical_compare_means.py	TC-CLIN-CMEAN-001..015	15/15 PASS
test_clinical_compare_props.py	TC-CLIN-CPROP-001..014	14/14 PASS

test_clinical_ancova.py	TC-CLIN-ANCOVA-001..013	13/13 PASS
test_clinical_logistic.py	TC-CLIN-LOGIT-001..013	13/13 PASS

6.3 jrc_clinical_ss_means — 14/14 PASS

TC ID	Description	Expected	Result
TC-CLIN-MEANS-001	Superiority valid inputs → exit 0, report header	Exit 0, 'Clinical sample size' + 'continuous endpoint'	PASS
TC-CLIN-MEANS-002	Superiority numeric (sd=10, $\delta=5$, power .80, α .05/2)	n treat = 63, n control = 63, n TOTAL = 126	PASS
TC-CLIN-MEANS-003	Sidedness reported with z-value	'2-sided' and z = 1.9600 in output	PASS
TC-CLIN-MEANS-004	Non-inferiority numeric (sd=8, M=4, power .90, α .025/1)	n treat = 85, n control = 85, n TOTAL = 170	PASS
TC-CLIN-MEANS-005	Dropout 0.15 inflation on TC-004 inputs	'ENROLL 100 + 100 = 200' in output	PASS
TC-CLIN-MEANS-006	Equivalence TOST numeric (sd=10, M=5, power .80)	n TOTAL = 138, 'equivalence (TOST)' in output	PASS
TC-CLIN-MEANS-007	Allocation 2:1 on TC-002 inputs	n treat = 95, n control = 48	PASS
TC-CLIN-MEANS-008	Sensitivity table over SD (x0.8...1.2)	5 scenario rows; totals 82 @ SD=8, 182 @ SD=12	PASS
TC-CLIN-MEANS-009	Superiority without --delta	Exit $\neq$ 0, 'requires --delta'	PASS
TC-CLIN-MEANS-010	Non-inferiority without --margin	Exit $\neq$ 0, 'requires --margin'	PASS
TC-CLIN-MEANS-011	Equivalence with $ \delta \geq \text{margin}$	Exit $\neq$ 0, ' $ \delta < \text{margin}$ '	PASS
TC-CLIN-MEANS-012	--power out of range (1.5)	Exit $\neq$ 0, '--power' in output	PASS
TC-CLIN-MEANS-013	Unknown flag	Exit $\neq$ 0, 'Unknown argument'	PASS
TC-CLIN-MEANS-014	Bypass protection — direct Rscript call fails	Exit $\neq$ 0, 'RENV_PATHS_ROOT' in output	PASS

6.4 jrc_clinical_ss_props — 11/11 PASS

TC ID	Description	Expected	Result
TC-CLIN-PROPS-001	Superiority valid inputs → exit 0, report header	Exit 0, 'Clinical sample size' + 'binary endpoint'	PASS
TC-CLIN-PROPS-002	Superiority numeric (pc .70, pt .85, power .80, α .05/2)	n treat = 118, n control = 118, n TOTAL = 236	PASS
TC-CLIN-PROPS-003	Non-inferiority numeric, default p-treat (pc .85, M .10)	n/arm = 201, n TOTAL = 402, 'assumed equal to control'	PASS
TC-CLIN-PROPS-004	Equivalence TOST numeric (pc .80, M .15, power .80)	n TOTAL = 244, 'equivalence (TOST)' in output	PASS
TC-CLIN-PROPS-005	Superiority without --p-treat	Exit $\neq$ 0, 'requires --p-treat'	PASS
TC-CLIN-PROPS-006	Superiority with p-treat = p-control	Exit $\neq$ 0, 'different from p-control'	PASS
TC-CLIN-PROPS-007	Non-inferiority without --margin	Exit $\neq$ 0, 'requires --margin'	PASS
TC-CLIN-PROPS-008	--p-control out of range (1.2)	Exit $\neq$ 0, '--p-control' in output	PASS
TC-CLIN-PROPS-009	Sensitivity table over p-control (x0.8...1.2)	5 scenario rows (0.64...0.96); assumed row marked	PASS

TC-CLIN-PROPS-010	Dropout 0.10 inflation on TC-003 inputs	'ENROLL 224 + 224 = 448' in output	PASS
TC-CLIN-PROPS-011	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

6.5 jrc_clinical_ss_survival — 11/11 PASS

TC ID	Description	Expected	Result
TC-CLIN-SURV-001	Superiority valid inputs → exit 0, report header	Exit 0, 'Clinical sample size' + 'time-to-event endpoint'	PASS
TC-CLIN-SURV-002	Superiority numeric (HR .70, power .80, α .05/2, PE .6)	Events = 247, n/arm = 206, n TOTAL = 412	PASS
TC-CLIN-SURV-003	Non-inferiority numeric (HR 1.0, M 1.3, α .025/1, PE .5)	Events = 457, n TOTAL = 914	PASS
TC-CLIN-SURV-004	--framework equivalence rejected	Exit ≠ 0, 'not offered'	PASS
TC-CLIN-SURV-005	Superiority with HR = 1	Exit ≠ 0, 'different from 1'	PASS
TC-CLIN-SURV-006	NI margin ≤ 1	Exit ≠ 0, '--margin' in output	PASS
TC-CLIN-SURV-007	NI with HR ≥ margin	Exit ≠ 0, '--hr < --margin'	PASS
TC-CLIN-SURV-008	Missing --event-prob	Exit ≠ 0, '--event-prob' in output	PASS
TC-CLIN-SURV-009	Sensitivity table over P(event) (×0.8...1.2)	5 scenario rows (0.48...0.72); assumed row marked	PASS
TC-CLIN-SURV-010	Dropout 0.20 inflation on TC-002 inputs	'ENROLL 258 + 258 = 516' in output	PASS
TC-CLIN-SURV-011	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

6.6 jrc_clinical_dx_accuracy — 20/20 PASS

TC ID	Description	Expected	Result
TC-CLIN-DXACC-001	Valid inputs → exit 0, report header	Exit 0, 'Diagnostic accuracy' + 'reference standard'	PASS
TC-CLIN-DXACC-002	2x2 contingency table from the fixture	TP 79, FP 6 / FN 16, TN 99; margins 95 / 105 / 200	PASS
TC-CLIN-DXACC-003	Sensitivity + Clopper-Pearson exact CI	0.8316 (79/95), 95% CI (0.7410, 0.9006)	PASS
TC-CLIN-DXACC-004	Specificity + Clopper-Pearson exact CI	0.9429 (99/105), 95% CI (0.8798, 0.9787)	PASS
TC-CLIN-DXACC-005	Overall accuracy + exact CI	0.8900 (178/200), 95% CI (0.8382, 0.9298)	PASS
TC-CLIN-DXACC-006	LR+ with log (Simel 1991) interval	LR+ = 14.5526, 95% CI (6.6563, 31.8163)	PASS
TC-CLIN-DXACC-007	LR- with log (Simel 1991) interval	LR- = 0.1786, 95% CI (0.1140, 0.2799)	PASS
TC-CLIN-DXACC-008	PPV/NPV at study prevalence, with validity caveat	PPV 0.9294, NPV 0.8609; 'study prevalence (0.4750)' + 'valid ONLY if'	PASS
TC-CLIN-DXACC-009	--ci wilson → Wilson score interval	Estimate 0.8316 unchanged; CI (0.7438, 0.8936), inside exact	PASS
TC-CLIN-DXACC-010	--prevalence 0.02 → Bayes-adjusted PPV/NPV	PPV (adj) 0.2290 (0.1117, 0.4637); NPV (adj) 0.9964	PASS
TC-CLIN-DXACC-011	--conf 0.90 narrows the interval	Estimate unchanged; both 90% bounds inside the 95% bounds	PASS
TC-CLIN-DXACC-012	--positive resolves non-standard labels	Exit 0; sensitivity 0.8000 (4/5), specificity 0.8000 (4/5)	PASS
TC-CLIN-DXACC-013	Rows with missing values dropped and reported	Exit 0; '4 evaluable' + '2 row(s) dropped'	PASS
TC-CLIN-DXACC-	Invalid --ci rejected	Exit ≠ 0, '--ci must be'	PASS

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TC-CLIN-DXACC-015	--prevalence out of range rejected	Exit ≠ 0, '--prevalence' named	PASS
TC-CLIN-DXACC-016	Missing required column rejected	Exit ≠ 0, 'Missing column(s): result'	PASS
TC-CLIN-DXACC-017	Duplicate subject id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-DXACC-018	Unrecognised labels rejected	Exit ≠ 0, 'does not recognise'	PASS
TC-CLIN-DXACC-019	No reference-negative subjects rejected	Exit ≠ 0, 'specificity is undefined'	PASS
TC-CLIN-DXACC-020	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

6.7 jrc_clinical_dx_ss — 14/14 PASS

TC ID	Description	Expected	Result
TC-CLIN-DXSS-001	Precision valid inputs → exit 0, report header	Exit 0, 'Clinical sample size' + 'diagnostic accuracy'	PASS
TC-CLIN-DXSS-002	Precision numeric — Buderer (1996) worked example	n ref+ = 139, n ref- = 73, N TOTAL = 1383	PASS
TC-CLIN-DXSS-003	Hypothesis numeric (goals .80/.90, power .80, α .025/1)	n ref+ = 108, n ref- = 239, N TOTAL = 1075	PASS
TC-CLIN-DXSS-004	Binding arm identified	'Binding arm : sensitivity (reference-positive)'	PASS
TC-CLIN-DXSS-005	Rarer prevalence raises enrolment, not arm size	n ref+ = 139 unchanged; N TOTAL = 2766 at prevalence 0.05	PASS
TC-CLIN-DXSS-006	Sensitivity table over prevalence (x0.5...2.0)	Exit 0; 5 rows, assumed row marked '<- assumed'	PASS
TC-CLIN-DXSS-007	Dropout 0.10 inflation on TC-003 inputs	'ENROLL 1195 subjects'	PASS
TC-CLIN-DXSS-008	Sidedness default differs by method	precision 'two-sided, z = 1.9600'; hypothesis '1-sided'	PASS
TC-CLIN-DXSS-009	Precision without --halfwidth	Exit ≠ 0, 'requires --halfwidth'	PASS
TC-CLIN-DXSS-010	Hypothesis without --sens-goal	Exit ≠ 0, 'requires --sens-goal'	PASS
TC-CLIN-DXSS-011	--sens-goal ≥ --sens-expected	Exit ≠ 0, '--sens-goal must be strictly less than'	PASS
TC-CLIN-DXSS-012	--prevalence out of range (1.2)	Exit ≠ 0, '--prevalence' named	PASS
TC-CLIN-DXSS-013	Invalid --method rejected	Exit ≠ 0, '--method must be one of'	PASS
TC-CLIN-DXSS-014	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

6.8 jrc_clinical_dx_roc — 17/17 PASS

TC ID	Description	Expected	Result
TC-CLIN-DXROC-001	Valid inputs → exit 0, report header	Exit 0, 'ROC analysis'; 80 ref+ / 120 ref-	PASS
TC-CLIN-DXROC-002	Two-panel PNG written to ~/Downloads/	*_jrc_clinical_dx_roc.png created during the run	PASS
TC-CLIN-DXROC-003	PUBLISHED ANCHOR — Hanley & McNeil (1982) Table 1	51 ref+ / 58 ref-; AUC = 0.8932 = published 0.893 (3 dp)	PASS
TC-CLIN-DXROC-004	AUC, DeLong SE and confidence interval	AUC 0.8268, SE 0.0294, 95% CI (0.7691, 0.8845)	PASS
TC-CLIN-DXROC-005	Tie-aware AUC kernel (ψ = 0.5 on ties)	12 ref+ / 14 ref-; AUC = 0.5952	PASS
TC-CLIN-DXROC-	Youden-optimal operating point	J = 0.5125, sensitivity 0.5875,	PASS

006		specificity 0.9250	
TC-CLIN-DXROC-007	Cutoff is an observed value with an explicit rule	'Cutoff : score >= 2.192 => positive'	PASS
TC-CLIN-DXROC-008	Test of AUC = 0.5	'H0: AUC = 0.5' and z = 11.1017	PASS
TC-CLIN-DXROC-009	--direction lower reflects AUC about 0.5	AUC 0.1732; 0.8268 + 0.1732 = 1 within 1e-9	PASS
TC-CLIN-DXROC-010	Degenerate cutoff reported as none, + AUC<0.5 warning	Exit 0; 'Cutoff : none', 'AUC < 0.5', no infinite threshold printed	PASS
TC-CLIN-DXROC-011	--ci-method logit interval	AUC unchanged 0.8268; 95% CI (0.7614, 0.8772)	PASS
TC-CLIN-DXROC-012	--conf 0.90 narrows the interval	AUC unchanged; 90% CI (0.7784, 0.8752), inside the 95% CI	PASS
TC-CLIN-DXROC-013	Invalid --direction rejected	Exit ≠ 0, '--direction must be'	PASS
TC-CLIN-DXROC-014	Missing required column rejected	Exit ≠ 0, 'Missing column(s): score'	PASS
TC-CLIN-DXROC-015	Duplicate subject id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-DXROC-016	No reference-negative subjects rejected	Exit ≠ 0, 'At least 2 reference-negative'	PASS
TC-CLIN-DXROC-017	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	PASS

6.9 jrc_clinical_km — 19/19 PASS

TC ID	Description	Expected	Result
TC-CLIN-KM-001	Valid inputs → exit 0, report header	Exit 0, 'Kaplan-Meier survival analysis', 23 subjects	PASS
TC-CLIN-KM-002	Two-panel PNG written	*_jrc_clinical_km.png created	PASS
TC-CLIN-KM-003	PUBLISHED ANCHOR — Miller (1981) medians	Maintained median 31, non-maintained 23	PASS
TC-CLIN-KM-004	PUBLISHED ANCHOR — Miller (1981) log-rank	$\chi^2 = 3.3964$ (pub. 3.4), df 1, p = 0.065 (0.07)	PASS
TC-CLIN-KM-005	Log-rank observed vs expected	Maint. 7/10.69, non-maint. 11/7.31	PASS
TC-CLIN-KM-006	Ungrouped overall median + CI	n 23, median 27, 95% CI (18, 45)	PASS
TC-CLIN-KM-007	Survival at time point	S(30) = 0.6136 and 0.2917	PASS
TC-CLIN-KM-008	Confidence level narrows median CI	90% CI (18, 43) inside 95% (18, 45)	PASS
TC-CLIN-KM-009	Rho-weighted (Peto) differs from log-rank	Peto $\chi^2 = 2.7793$, ≠ log-rank 3.3964	PASS
TC-CLIN-KM-010	Event-label override	Exit 0, 6 subjects (km_yneno, --event-positive yes)	PASS
TC-CLIN-KM-011	Yes/no labels auto-detected	Exit 0, 3 events	PASS
TC-CLIN-KM-012	Median CI reported NA when not reached	Maintained median CI upper = NA	PASS
TC-CLIN-KM-013	Missing required column rejected	Exit ≠ 0, 'Missing column(s)'	PASS
TC-CLIN-KM-014	Duplicate id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-KM-015	All-censored data rejected	Exit ≠ 0, 'No events observed'	PASS
TC-CLIN-KM-016	Single-group comparison rejected	Exit ≠ 0, 'only one distinct value'	PASS
TC-CLIN-KM-017	Unrecognised event labels rejected	Exit ≠ 0, 'does not recognise'	PASS
TC-CLIN-KM-018	Non-positive time rejected	Exit ≠ 0, 'strictly positive'	PASS
TC-CLIN-KM-019	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS

6.10 jrc_clinical_coxph — 17/17 PASS

TC ID	Description	Expected	Result
TC-CLIN-COX-001	Valid inputs → exit 0, report header	Exit 0, 'Cox proportional-hazards', 228 subj / 165 events	PASS
TC-CLIN-COX-002	Schoenfeld PNG written	*_jrc_clinical_coxph.png created	PASS
TC-CLIN-COX-003	PUBLISHED ANCHOR — Therneau sex HR	sex HR 0.5986, 95% CI (0.4311, 0.8311)	PASS
TC-CLIN-COX-004	PUBLISHED ANCHOR — Therneau age HR	age HR 1.0172, 95% CI (0.9990, 1.0357)	PASS
TC-CLIN-COX-005	Harrell's concordance	C = 0.6029	PASS
TC-CLIN-COX-006	Global LR and Wald tests	LR χ^2 14.1231, Wald χ^2 13.4700	PASS
TC-CLIN-COX-007	PH assumption OK when it holds	'PH assumption OK', global p = 0.2502	PASS
TC-CLIN-COX-008	PUBLISHED ANCHOR — PH violation flagged	'PH assumption VIOLATED', p = 0.000316 (veteran/karno)	PASS
TC-CLIN-COX-009	Factor covariate	sex_labelmale HR 1.7007 (= 1/0.5986)	PASS
TC-CLIN-COX-010	Missing covariate rows dropped	227 fitted, 'Dropped rows : 1'	PASS
TC-CLIN-COX-011	Breslow ties differ from Efron	sex HR 0.5990 (breslow) ≠ 0.5986 (efron)	PASS
TC-CLIN-COX-012	Confidence level narrows HR CI	90% CI (0.4545, 0.7884) inside 95%	PASS
TC-CLIN-COX-013	Missing covariate column rejected	Exit ≠ 0, 'Missing column(s)'	PASS
TC-CLIN-COX-014	--covariates required	Exit ≠ 0, '--covariates is required'	PASS
TC-CLIN-COX-015	Duplicate id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-COX-016	Fewer than two events rejected	Exit ≠ 0, 'Fewer than 2 events'	PASS
TC-CLIN-COX-017	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS

6.11 jrc_clinical_dx_compare — 18/18 PASS

TC ID	Description	Expected	Result
TC-CLIN-DXCMP-001	Valid inputs → exit 0, report header	Exit 0, 'Paired ROC comparison', 70 ref+ / 100 ref-	PASS
TC-CLIN-DXCMP-002	Two-panel PNG written	*_jrc_clinical_dx_compare.png created	PASS
TC-CLIN-DXCMP-003	ORACLE ANCHOR — per-test AUCs (pROC)	AUC A 0.8681 (0.8138,0.9225); AUC B 0.6670 (0.5851,0.7489)	PASS
TC-CLIN-DXCMP-004	Difference A-B with CI	+0.2011 (95% CI +0.1148, +0.2875)	PASS
TC-CLIN-DXCMP-005	Paired DeLong test	z 4.5640, p 5.019e-06 (bit-identical to pROC roc.test)	PASS
TC-CLIN-DXCMP-006	Between-AUC correlation	0.2468	PASS
TC-CLIN-DXCMP-007	Significance statement	'test_a has the higher AUC ... significant'	PASS
TC-CLIN-DXCMP-008	--conf 0.90 narrows difference CI	90% CI (+0.1287, +0.2736)	PASS
TC-CLIN-DXCMP-009	Logit per-test CI differs, in (0,1)	AUC A logit CI (0.8037, 0.9137)	PASS
TC-CLIN-DXCMP-010	Direction lower reflects both, flips z	AUC A 0.1319, B 0.3330; z -4.5640	PASS
TC-CLIN-DXCMP-011	--tests names the two columns	'A = test_a vs B = test_b'	PASS
TC-CLIN-DXCMP-012	Two columns auto-inferred	test_a vs test_b without --tests	PASS
TC-CLIN-DXCMP-013	Ambiguous columns rejected	Exit ≠ 0, 'Could not infer the two test columns'	PASS
TC-CLIN-DXCMP-	Named column absent rejected	Exit ≠ 0, 'Test column(s) not found:	PASS

014		nosuchcol'	
TC-CLIN-DXCMP-015	Missing reference rejected	Exit $\neq$ 0, 'Missing column(s): reference'	PASS
TC-CLIN-DXCMP-016	Duplicate id rejected	Exit $\neq$ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-DXCMP-017	No reference-negative rejected	Exit $\neq$ 0, 'At least 2 reference-negative'	PASS
TC-CLIN-DXCMP-018	Bypass protection — direct Rscript call fails	Exit $\neq$ 0, 'RENV_PATHS_ROOT'	PASS

6.12 jrc_clinical_compare_means — 15/15 PASS

TC ID	Description	Expected	Result
TC-CLIN-CMEAN-001	Valid inputs → exit 0, header	Exit 0, 20 subjects	PASS
TC-CLIN-CMEAN-002	ANCHOR — Student (1908) Welch t	t=-1.8608 df=17.78 p=0.0794	PASS
TC-CLIN-CMEAN-003	Mean difference + CI	-1.5800 (95% CI -3.3655, +0.2055)	PASS
TC-CLIN-CMEAN-004	Effect size	Cohen d=-0.8322, Hedges g=-0.7970	PASS
TC-CLIN-CMEAN-005	Mann-Whitney	W=25.5 p=0.06582	PASS
TC-CLIN-CMEAN-006	--test student (equal var)	t=-1.8608 df=18 p=0.07919	PASS
TC-CLIN-CMEAN-007	--conf 0.90 narrows CI	90% CI (-3.0534, -0.1066)	PASS
TC-CLIN-CMEAN-008	Per-group normality note	'Shapiro p =' per group	PASS
TC-CLIN-CMEAN-009	ANCHOR — PlantGrowth ANOVA	F=4.8461 df 2,27 p=0.0159	PASS
TC-CLIN-CMEAN-010	Missing column rejected	Exit $\neq$ 0, 'Missing column(s)'	PASS
TC-CLIN-CMEAN-011	Duplicate id rejected	Exit $\neq$ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-CMEAN-012	Single group rejected	Exit $\neq$ 0, 'At least two groups'	PASS
TC-CLIN-CMEAN-013	Group < 2 obs rejected	Exit $\neq$ 0, 'fewer than 2 observations'	PASS
TC-CLIN-CMEAN-014	Bad --test rejected	Exit $\neq$ 0, '--test must be'	PASS
TC-CLIN-CMEAN-015	Bypass protection	Exit $\neq$ 0, 'RENV_PATHS_ROOT'	PASS

6.13 jrc_clinical_compare_props — 14/14 PASS

TC ID	Description	Expected	Result
TC-CLIN-CPROP-001	Valid inputs → exit 0	Exit 0, contingency table	PASS
TC-CLIN-CPROP-002	ANCHOR — chi-square (Yates)	X2=5.6201 df=1 p=0.01776	PASS
TC-CLIN-CPROP-003	Fisher exact	p=0.01715	PASS
TC-CLIN-CPROP-004	ANCHOR — risk difference	-0.1500 (95% CI -0.2639, -0.0361)	PASS
TC-CLIN-CPROP-	ANCHOR — risk ratio	0.5000 (95% CI 0.2872, 0.8704)	PASS

005			
TC-CLIN-CPROP-006	ANCHOR — odds ratio	0.4118 (95% CI 0.2053, 0.8258)	PASS
TC-CLIN-CPROP-007	--no-correction	X2=6.4516 p=0.01109	PASS
TC-CLIN-CPROP-008	--reference flips OR	OR 2.4286 (1.2110, 4.8703)	PASS
TC-CLIN-CPROP-009	Event auto-detected	event = 'yes'	PASS
TC-CLIN-CPROP-010	Missing column rejected	Exit ≠ 0, 'Missing column(s)'	PASS
TC-CLIN-CPROP-011	Duplicate id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-CPROP-012	Single group rejected	Exit ≠ 0, 'At least two groups'	PASS
TC-CLIN-CPROP-013	Unknown --event rejected	Exit ≠ 0, 'not an outcome level'	PASS
TC-CLIN-CPROP-014	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS

6.14 jrc_clinical_ancova — 13/13 PASS

TC ID	Description	Expected	Result
TC-CLIN-ANCOVA-001	Valid inputs → exit 0	Exit 0, 72 subjects	PASS
TC-CLIN-ANCOVA-002	ANCHOR — anorexia Type III group	group F=7.8680 df 2,68 p=0.0008	PASS
TC-CLIN-ANCOVA-003	Type III covariate test	prewt F=7.2660 df 1,68 p=0.00885	PASS
TC-CLIN-ANCOVA-004	ANCHOR — adjusted LS means	CBT 85.5743 Cont 81.4773 FT 90.1374	PASS
TC-CLIN-ANCOVA-005	Tukey pairwise contrast	Cont-FT -8.6601 p=0.0005484	PASS
TC-CLIN-ANCOVA-006	Covariate coefficient	prewt +0.4345 (SE 0.1612)	PASS
TC-CLIN-ANCOVA-007	--conf 0.90 narrows CI	CBT 90% CI (83.4121, 87.7365)	PASS
TC-CLIN-ANCOVA-008	Missing column rejected	Exit ≠ 0, 'Missing column(s)'	PASS
TC-CLIN-ANCOVA-009	--covariates required	Exit ≠ 0, '--covariates is required'	PASS
TC-CLIN-ANCOVA-010	Missing covariate rejected	Exit ≠ 0, 'Missing column(s)'	PASS
TC-CLIN-ANCOVA-011	Duplicate id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-ANCOVA-012	Single group rejected	Exit ≠ 0, 'at least two levels'	PASS
TC-CLIN-ANCOVA-013	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS

6.15 jrc_clinical_logistic — 13/13 PASS

TC ID	Description	Expected	Result
TC-CLIN-LOGIT-001	Valid inputs → exit 0	Exit 0, 189 subjects	PASS
TC-CLIN-LOGIT-002	ANCHOR — birthwt smoke OR	OR 1.9557 (1.0326, 3.7042)	PASS
TC-CLIN-LOGIT-003	Continuous predictor OR	lwt OR 0.9879 (0.9761, 0.9999)	PASS

TC-CLIN-LOGIT-004	Non-significant predictor	age OR 0.9618 p=0.2334	PASS
TC-CLIN-LOGIT-005	Global LR test	chisq 11.7926 df 3 p=0.008128	PASS
TC-CLIN-LOGIT-006	Model fit (AIC)	AIC 230.88	PASS
TC-CLIN-LOGIT-007	In-sample model AUC	AUC 0.6531	PASS
TC-CLIN-LOGIT-008	--conf 0.90 narrows OR CI	smoke OR (1.1442, 3.3427)	PASS
TC-CLIN-LOGIT-009	Missing column rejected	Exit ≠ 0, 'Missing column(s)'	PASS
TC-CLIN-LOGIT-010	--predictors required	Exit ≠ 0, '--predictors is required'	PASS
TC-CLIN-LOGIT-011	Non-binary outcome rejected	Exit ≠ 0, 'exactly two levels'	PASS
TC-CLIN-LOGIT-012	Duplicate id rejected	Exit ≠ 0, 'Duplicate id(s) found'	PASS
TC-CLIN-LOGIT-013	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	PASS

7. Deviations

No deviations were identified during OQ execution. All 196 test cases passed.

8. Requirements Traceability Matrix

User requirements UR-CLIN-001 ... UR-CLIN-196 map 1:1, in order, to test cases TC-CLIN-MEANS-001..014, TC-CLIN-PROPS-001..011, TC-CLIN-SURV-001..011, TC-CLIN-DXACC-001..020, TC-CLIN-DXSS-001..014 and TC-CLIN-DXROC-001..017 (see JR-VP-CLIN-001 Rev 2 for the full matrix). UR-CLIN-001 ... UR-CLIN-036 are unchanged from Rev 1 and retain their original numbering; Module 2 appends UR-CLIN-037 ... UR-CLIN-087. Grouped by capability:

Requirement group	Capability	Test Cases
UR-CLIN-001..008	jrc_clinical_ss_means — frameworks, numerics, allocation, dropout, sensitivity	TC-CLIN-MEANS-001..008
UR-CLIN-009..013	jrc_clinical_ss_means — input validation	TC-CLIN-MEANS-009..013
UR-CLIN-014	jrc_clinical_ss_means — bypass protection	TC-CLIN-MEANS-014
UR-CLIN-015..018	jrc_clinical_ss_props — frameworks and numerics	TC-CLIN-PROPS-001..004
UR-CLIN-019..024	jrc_clinical_ss_props — input validation, sensitivity, dropout	TC-CLIN-PROPS-005..010
UR-CLIN-025	jrc_clinical_ss_props — bypass protection	TC-CLIN-PROPS-011
UR-CLIN-026..028	jrc_clinical_ss_survival — frameworks and numerics (events → subjects)	TC-CLIN-SURV-001..003
UR-CLIN-029..035	jrc_clinical_ss_survival — input validation, sensitivity, dropout	TC-CLIN-SURV-004..010
UR-CLIN-036	jrc_clinical_ss_survival — bypass protection	TC-CLIN-SURV-011
UR-CLIN-037..044	jrc_clinical_dx_accuracy — 2x2, sens/spec/accuracy, LR± with intervals	TC-CLIN-DXACC-001..008
UR-CLIN-045..049	jrc_clinical_dx_accuracy — interval methods, Bayes-adjusted PPV/NPV, labels, missing data	TC-CLIN-DXACC-009..013
UR-CLIN-050..055	jrc_clinical_dx_accuracy — input validation	TC-CLIN-DXACC-014..019
UR-CLIN-056	jrc_clinical_dx_accuracy — bypass protection	TC-CLIN-DXACC-020
UR-CLIN-057..064	jrc_clinical_dx_ss — precision and hypothesis sizing, prevalence, dropout, sensitivity	TC-CLIN-DXSS-001..008
UR-CLIN-065..069	jrc_clinical_dx_ss — input validation	TC-CLIN-DXSS-009..013
UR-CLIN-070	jrc_clinical_dx_ss — bypass protection	TC-CLIN-DXSS-014
UR-CLIN-071..073	jrc_clinical_dx_roc — report, plot output, PUBLISHED AUC anchor (Hanley & McNeil 1982)	TC-CLIN-DXROC-001..003
UR-CLIN-074..082	jrc_clinical_dx_roc — AUC/DeLong interval, tie kernel, Youden cutoff, direction, degenerate cutoff, interval options	TC-CLIN-DXROC-004..012

UR-CLIN-083..086	jrc_clinical_dx_roc — input validation	TC-CLIN-DXROC-013..016
UR-CLIN-087	jrc_clinical_dx_roc — bypass protection	TC-CLIN-DXROC-017

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Clinical Study Design Module v1.0 is complete.

Result: PASS

All 196 automated test cases specified in JR-VP-CLIN-001 v1.0 were executed on 2026-07-20 and 196 passed. Results are derived from the OQ evidence file listed below.

- jrc_clinical_ss_means — two-arm sample size, continuous endpoint (superiority / non-inferiority / equivalence)
- jrc_clinical_ss_props — two-arm sample size, binary endpoint, risk difference (superiority / non-inferiority / equivalence)
- jrc_clinical_ss_survival — two-arm sample size, time-to-event endpoint, log-rank (superiority / non-inferiority)

Evidence of test execution is archived at: clinical_oq_execution_20260720T104013.txt

The Clinical Study Design module is released as version 1.0 of repos/clinical/. Any subsequent change to the scripts, wrappers, or help files within repos/clinical/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.